MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE USAF 6233

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pulsed, tunable frequency range 9280 to 9345 MHz, integral magnet, rated peak power output 7 kW.

ABSOLUTE RATINGS:

Parameter:	Ef	epy	ib	pi	Pi	tk	Du
Unit:	V	kv	a	kw	W	sec	----
Maximum:	7.0	7.0	5	35	75	180	.003
Minimum:	---	---	--	--	--	---	----

ABSOLUTE RATINGS:

Parameter:	tp	TA	Tuner Torque
Unit:	μs	°C	in-oz
Maximum:	1.0	±150	150
Minimum:	---	-55	10

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1
Weight:	4 lbs approximately
Cooling:	Force air (see note 2)
Cathode:	Unipotential oxide coated

TEST CONDITIONS:

Parameter:	Ef	F	tp	Du	trv	Ib	TA	VSWR
Unit:	V	MHz	μs	--	μs	mAdc	°C	----
Tolerance:	---	1.5	±0.05	--	--	----	--	----
Test 1:	6.3	9310	0.5	0.003	0.2 (max)	11.5	125 (max)	1.05
Test 2:	6.3	9310	0.5	0.001	0.2 (max)	3.8	125	1.05
Test 3:	6.3	9310	0.5	0.002	0.2 (max)	7.7	125	1.05

GENERAL:

Qualification - Not required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	TEST	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1						
1301	Heater current	Ef = 6.3 V	-	I_f	0.90	1.10	A
4223	Mechanical tuning range	See note 4	-				
		Upper limit		F	9345	---	MHz
		Lower limit		F	---	9280	MHz
4306	Pulse voltages		1,2	epy	5.5	6.0	kv
4250	Power output		1	Po	21	---	W
4308	RF bandwidth		1,2	BW	---	5	MHz
4315	Starting stability	VSWR = 1.50 See note 3	2	MP	---	0.5	%
	Quality conformance inspection, part 2						
1031	High-frequency vibration	No voltages	-	---	---	---	---
1031	Low-frequency vibration	No voltages	-	---	---	---	---
4223	Tuner operating torque	See note 6	-	Torque	---	10	oz-in
4310	Frequency pulling figure		1	ΔF	---	20	MHz
	Quality conformance inspection, part 3						
4551	Life test	Group D	3	t	500	---	hrs
----	Life test end points						
4250	Power output		1	Po	17	---	W
4308	RF bandwidth		1,2	BW	---	6.0	MHz
	Periodic check tests	See note 1					
1047	Low-temperature operation	tk = 180 sec (max)	1	PO	21	---	W
4027	Temperature coefficient		1	ΔF/ΔT	---	0.25	MHz/°C
1042	Shock		-	---	---	---	---

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NOTES:

1. If these tests have been previously performed by the tube manufacturer, and if the material, manufacturing process, and test requirements are the same as when the test(s) were performed, the subject test(s) need not be performed.

2. Cooling air shall be supplied to the tube in the amount which will maintain the anode at 125°C or below, up to the maximum duty cycle.

3. The tube shall be considered stable when the number of missing pulses is a maximum of 0.5 percent of the total number of pulses during a five minute test period. A missing pulse is defined as any pulse containing less than 90 percent of the energy at the pi mode operating frequency contained in a normal pulse.

4. At an anode temperature of 125°C, the tube shall be capable of tuning the entire range of 9280 to 8345 MHz in 2.33 turns of the external tuner shaft.

5. It is necessary to lower the heater voltage to 4.5 volts for operation of 0.002 duty or greater.

6. Tuner torque shall be less than 10 oz.-in over a temperature range of -30°C to +125°C.

Custodian:
Air Force - 80

Preparing activity:
Air Force - 80

Review activities:
Air Force - 11, 84

(Project 5960-F852)

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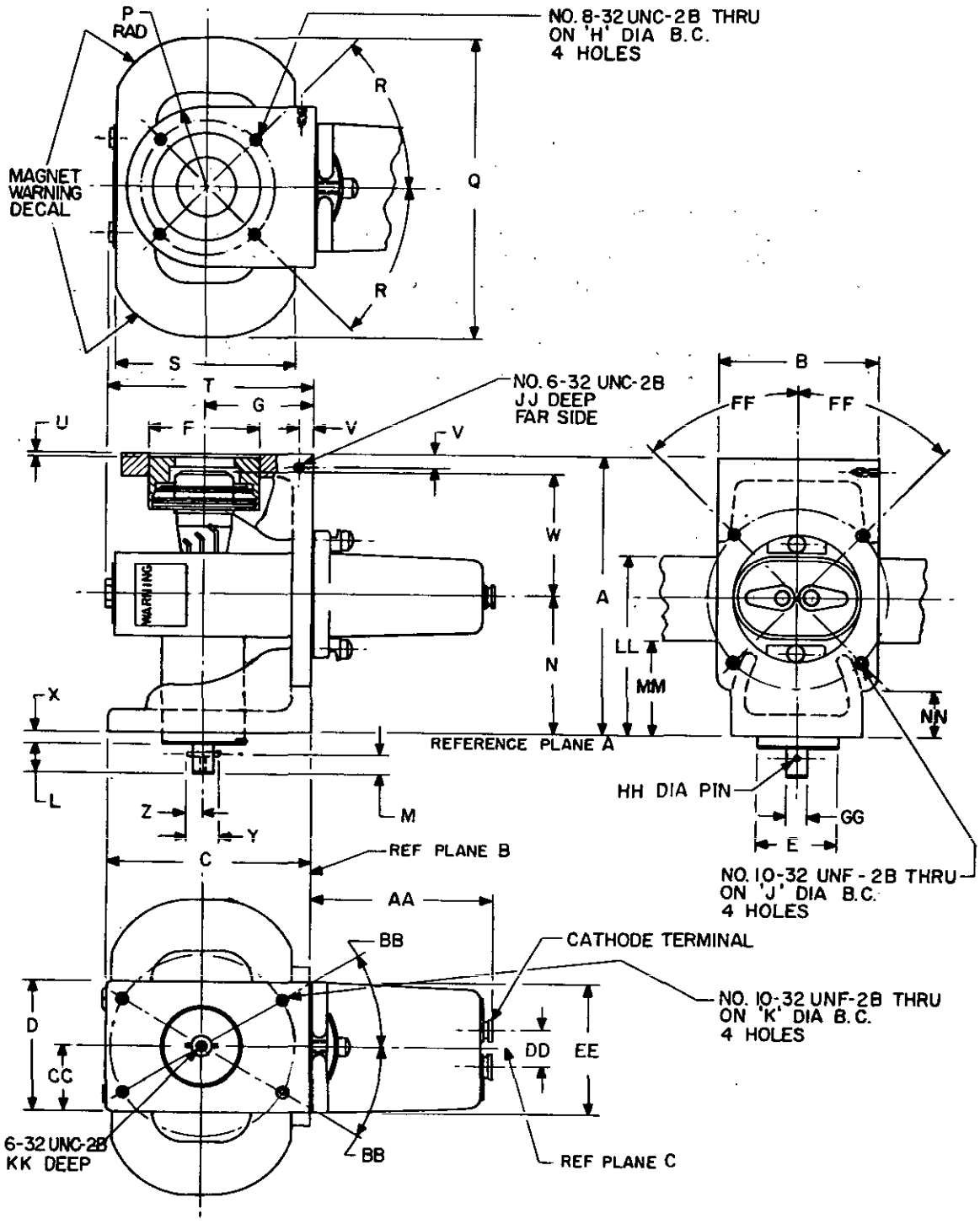


FIGURE 1 Outline of type USAF 6233

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DIMENSIONS				
LTR	INCHES		MM	
	MIN	MAX	MIN	MAX
QUALITY CONFORMANCE INSPECTION, PART 1				
A	4.623	4.633	117.42	117.68
B	–	2.687	–	68.25
C	–	3.312	–	84.12
D	–	2.187	–	55.55
E	1.3105	1.3115	33.29	33.31
F	1.750	1.755	44.45	44.58
G	1.745	1.755	44.32	44.58
H	2.182	2.192	55.42	55.68
J	2.870	2.880	72.90	73.15
K	2.995	3.005	76.07	76.33
L	.406	.453	10.31	11.51
M	.312	.343	7.94	8.71
N	2.290	2.310	58.17	58.67

LTR	INCHES		MM	
	MIN	MAX	MIN	MAX
QUALITY CONFORMANCE INSPECTION, PART 2				
P	–	1.312	–	33.32
Q	–	5.375	–	136.53
R	44° 40'	45° 20'		
S	–	3.000	–	76.20
T	–	3.500	–	88.90
U	.054	.070	1.37	1.78
X	.177	.197	4.50	5.00
Y	.406	.453	10.31	11.51
Z	.213	.223	5.41	5.66
AA	–	3.047	–	77.39
BB	29° 45'	30° 15'		
DD	.490	.510	12.45	12.95
FF	44° 45'	45° 15'		
GG	.3116	.3121	7.91	7.93
LL	–	3.120	–	79.25
MM	1.480	–	37.59	–

NOMINAL DIMENSIONS		
LTR	INCHES	MM
V	.187	4.75
W	2.077	52.76
CC	1.093	27.76
EE	2.062	52.37
HH	.062	1.57
JJ	.250	6.35
KK	.312	7.92
NN	.750	19.05

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